

Automated Quantifiable Assessments of Sensorimotor Function Using an Instrumented Fragile Object

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Abstract— Accurate assessment of hand dexterity plays a critical role in informing rehabilitation and care of upper-limb hemiparetic stroke patients. Common upper-limb assessments, such as the Box and Blocks Test and Nine Hole Peg Test, primarily evaluate gross motor function in terms of speed. These assessments neglect an individual's ability to finely regulate grip force, which is critical in activities of daily living, such as manipulating fragile objects. Here we present the Electronic Grip Gauge (EGG), an instrumented fragile object that assesses both gross and fine motor function. Embedded with a load cell, accelerometer, and Hall-effect sensor, the EGG measures grip force, acceleration, and relative position (via magnetic fields) in real time. The EGG can emit an audible “break” sound when the applied grip force exceeds a threshold. The number of breaks, transfer duration, and applied forces are automatically logged in real-time. Using the EGG, we evaluated sensorimotor function in implicit grasping and gentle grasping for the non-paretic and paretic hands of 3 hemiparetic stroke patients. For all participants, the paretic hand took longer to transfer the EGG during implicit grasping. For 2 of 3 participants, grip forces were significantly greater for the paretic hand during gentle grasping. Differences in implicit grasping forces were unique to each participant. This work constitutes an important step towards more widespread and quantitative measures of sensorimotor function, which may ultimately lead to improved personalized rehabilitation and better patient outcomes.

I. INTRODUCTION

Worldwide by 2030, the population burdened by a stroke-related disability is expected to reach 70 million [1]. In the United States alone, a new stroke occurs on average every 40 seconds, leading to approximately 800,000 Americans suffering a stroke each year [2]. About two-thirds of those who suffer a stroke survive beyond the three-year mark [3] and require ongoing neurorehabilitation. Upper-limb hemiparesis is the most prevalent and lasting disability resulting from stroke, with more than 80% of stroke patients experiencing this condition acutely and 40% experiencing it chronically [4]. Upper-limb hemiparesis is associated with weakness, numbness, pain, and/or paralysis in the hand and arm on one side of the body. These sensorimotor impairments can substantially impact a patient's quality of life.

The sensorimotor deficits resulting from a stroke vary widely among patients, and therefore, personalized rehabilitation is recommended. Accurate assessment of a stroke patient's abilities plays a critical role in guiding

personalized rehabilitation [5]. Common tests of motor function, such as the Box and Blocks Test (BBT) [6] and Nine Hole Peg Test (9HPT) [7], provide a unilateral measure of gross motor function and manual dexterity, respectively. The BBT and 9HPT, developed in the 1950s through 1970s, are simple to perform but provide little quantitative data and do not directly measure finer sensorimotor function, such as the ability to regulate grip force to manipulate fragile objects. Similarly, existing sensory assessments, such as the Two-Point Discrimination Test (2PDT) [8] and the Semmes-Weinstein monofilament test (SWM) [8], are performed in a passive, open-loop manner that ignores the sensorimotor integration necessary for fine sensorimotor dexterity.

To assess fine sensorimotor function, variants on the BBT have been developed which introduce a “break” threshold to simulate handling a fragile object such as an egg. Here the user must maintain their grip force above the minimum required to pick up the object while keeping it below the force that would cause it to break [9]. Various implementations of this fragile-object task have been developed, including using metal plates separated by a weak magnetic field [10]–[12], paper blocks held together loosely with toothpicks [13], and 3D-printed blocks with embedded magnets [14]–[16]. A more recent adaption involves a 3D-printed block integrated with a strain gauge and accelerometer for precise measurement of grip force and load force exerted on the object [9], [17], [18].

We previously introduced our own instantiation of a fragile object, dubbed the Electronic Grip Gauge (EGG), involving a 3D-printed block embedded with a load cell and an accelerometer [9]. Here we build upon prior work by extending the fragile-object task to stroke patients and by introducing a novel sensing modality to automate data collection. By providing preliminary evidence of quantifiable sensorimotor function in stroke patients and by demonstrating the feasibility of automatically capturing data in real-time, we hope to translate the fragile-object task into a standard and widely used clinical assessment of fine sensorimotor function.

II. METHODS

A. Device Design

The EGG design detailed here is a further evolution of our previous work [9]. Briefly, the EGG consists of a 100-lbf load cell (TE Connectivity Measurement Specialties), triple-axis accelerometer (ADXL335BCPZ-RL7), Hall-effect sensor

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(SS49E), custom designed printed circuit board (EGG 2.0 Shield), and wireless microcontroller (Adafruit Esp 32 Feather V2), housed within a 3D-printed polylactic (PLA) shell (Fig. 1). Sensor readings from the load cell, accelerometer, and Hall-effect sensor are captured by the microcontroller and streamed to a computer wirelessly at 100 Hz using a User Datagram Protocol. The load cell, accelerometer, and Hall-effect sensor transmit force, x -axis, y -axis, z -axis acceleration, and magnetic field strength and direction, respectively. Notable changes from [9] include the addition of a custom printed circuit board, integrated charging circuit via the Feather microcontroller, and the Hall-effect sensor, which allows for automatic data segmentation and monitoring of the relative position of the EGG.

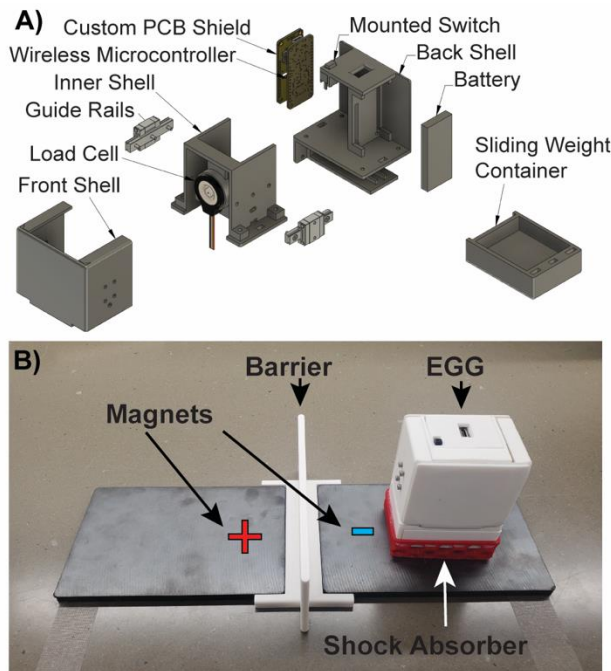


Figure 1. Electronic Grip Gauge (EGG). (A) Exploded view of computer-aided design of the EGG. (B) Image of the EGG and a 3D-printed barrier separating two large magnets of opposite polarity. Participants were tasked with transferring the EGG over the barrier from one magnet to the other. The embedded loadcell monitors the grip force exerted on the EGG. The EGG can be configured to simulate a fragile object by emitting a “break” sound if the applied grip force exceeds a specified threshold. The embedded Hall-effect sensor can be used to detect the vertical position of the EGG (via the magnitude of the magnetic field strength) and the horizontal position of the EGG relative to the barrier (via the polarity of the magnetic field).

The base of the EGG can be filled with lead sheets to vary the weight of the object between 187 g and 351 g. The assembled EGG measures 66 x 46.5 x 77.1 mm and is appropriately sized to be grasped by paretic and non-paretic hands. An embedded battery (3.7V 800mAh) provides a battery life of more than 3 hours. The EGG can be turned on and off by a switch mounted on the top of the EGG. The EGG can also be reprogrammed and recharged via a single USB-C port located on the top of the EGG.

B. Participants and Standard Assessments

Three stroke participants with chronic hemiparesis

volunteered for this study. Participants were between the ages of 22 and 27 (100% female and > 9 months since their most recent stroke). Informed consent and experimental protocols were carried out in accordance with the University of Utah Institutional Review Board.

Participants rated their own motor and sensory function for both their non-paretic and paretic hands using an 11-point scale with 0 representing no function at all and 10 representing completely normal function (Table 1). The Modified Ashworth Scale (MAS) was also performed on the digits as a measure of hand spasticity. The MAS is scored on a 6-point scale from range from 0 to 4. A score of 0 indicates normal or no increase in muscle tone during flexion/extension whereas a score of 4 represents static rigidity with little to no flexion or extension possible. Participants also completed the BBT, 9HPT, 2PDT, SWMT to assess baseline motor and sensory function, following the procedures outlined in [19]. Both 2PDT and SWMT were performed on the distal portion of the index finger. Participant P1 was unable to complete the 9HPT and was found to have no sensory function during 2PDT and SWMT in their paretic hand. Participant P3 was unable to complete the 9HPT. For BBT, greater than 60 blocks transferred in a minute is considered normal [6]. For 9HPT, roughly 16 ± 2 seconds to transfer and replace all 9 pegs is considered normal [7]. For 2PDT, normal sensory function falls within 2 to 5 millimeters [20]. For SWMT, target forces 0.008 – 0.07 gm are considered normal and 0.16 – 0.4 gm are considered diminished light touch [21].

Table 1. Participant Sensorimotor Function

Participant	P1	P2	P3
Paretic Hand (yes/no)	✓	✓	✓
Dominant Hand Paretic (yes/no)	X	✓	✓
Sensory Survey (0-10)	10	4	10
Motor Survey (0-10)	10	4	10
MAS (0-4)	0	0	2
BBT (# blocks)	33	1	56
9HPT (seconds)	23	-	26
2PDT (mm)	3	-	3
SWMT (Target Force gm)	.07	-	0.16

C. Experimental Tasks

Participants completed two test conditions with the EGG: 1) an implicit grasping task, and 2) a gentle grasping task. In the implicit grasping condition, participants were instructed to grasp the EGG and transfer it over a 2”-tall 3D-printed barrier as quickly as possible 20 times (Fig. 1B). Participants first completed the task with their non-paretic hand, and then completed the task with their paretic hand. After transferring the object over the barrier, the participant would release the object completely and then repeat the transfer in the opposite direction. During each transfer, grip force and transfer duration were recorded. Because no instruction was provided to the participant on how to grasp the EGG, this condition measured a user’s implicit grasping force, similar to the grasping relative index of performance [22], and maximal transfer rate, similar to the BBT.

For the gentle grasping condition, a break threshold was introduced, such that the EGG would “break” and emit an

audible sound if the participant's grip force exceeded 9.5 N. Participants were again instructed to pick up the EGG and transfer it as quickly as possible. However, this time, they were also instructed to minimize their grip force so as not to hear the EGG "break." Task difficulty can be adjusted by decreasing the ratio of the break threshold relative to the object weight. In this study, the EGG weighed 351 grams and was set to a break threshold of 9.5N, such that the ratio of break force to weight was 0.027 N/g, as used in [9]. Other studies have used ratios of 0.032 N/g, 0.02 N/g, 0.015 N/g, and 1.34 N/g [23] [24], [25] [26], [27]. During this task, transfer grip force, transfer duration, and number of breaks were recorded. This gentle grasping condition provided a measure of the users innate ability to regulate their fine motor function during an object transfer task, an exercise common in many activities of daily living and one that hemiparetic stroke patients often experience degrees of difficulty with on their paretic side [28].

D. Automatic Performance Metrics

The EGG uses a Hall-effect sensor to detect local magnetic fields (Fig. 1B), but the EGG itself is not magnetic and is not impacted by local magnetic fields. The magnetic field data was zero-centered and filtered using a 10-sample moving average. The absolute magnitude of the magnetic field can be used to determine the relative vertical position of the EGG relative to the magnetic surface. The polarity of the magnetic field can be used to determine the relative horizontal position of the EGG relative to the 3D-printed barrier separating positive and negative magnets.

Attempted transfers were first identified when the value of the magnetic field strength dropped below the +0.015-T threshold, indicating a lift-off. The attempted transfer was labeled as a successful transfer if the magnetic field strength, then switched polarity dropping below the -0.015-T threshold, thereby indicating it was placed on the opposite side of the 3D-printed barrier (Fig. 2A). If the magnetic field returned to the same polarity, the attempted transfer was instead labeled as a failed transfer since the EGG never crossed the 3D-printed barrier (Fig. 2B).

After identifying successful transfers via the magnetic field, the duration of the transfer was identified as the time that occurred between crossing below the +0.015-T and -0.015-T thresholds. The peak grip force was then measured as the maximum grip force within the transfer duration window plus an additional 0.66 seconds (20 samples) before and after. The 0.66 seconds before and after were added to capture grip force data that occurred before the EGG had lifted off the magnet and after the EGG had been placed back on the magnet. If at any point the grip force recorded during the transfer exceeded the break threshold of 9.5 N, the transfer was labeled a failed grasp, indicating a "break" (Fig. 2C). The methods outlined above were also applied when transferring in the opposite direction (ascending from negative to positive polarity).

The transfer force was calculated as $F = m \cdot (|a| - g)$, where F is the transfer force, m is the mass of the EGG (351 g), g is gravity, and $|a|$ is the magnitude of the acceleration vector defined as $|a| = \sqrt{x^2 + y^2 + z^2}$, where x , y , and z are the x -axis, y -axis, and z -axis acceleration vectors respectively.

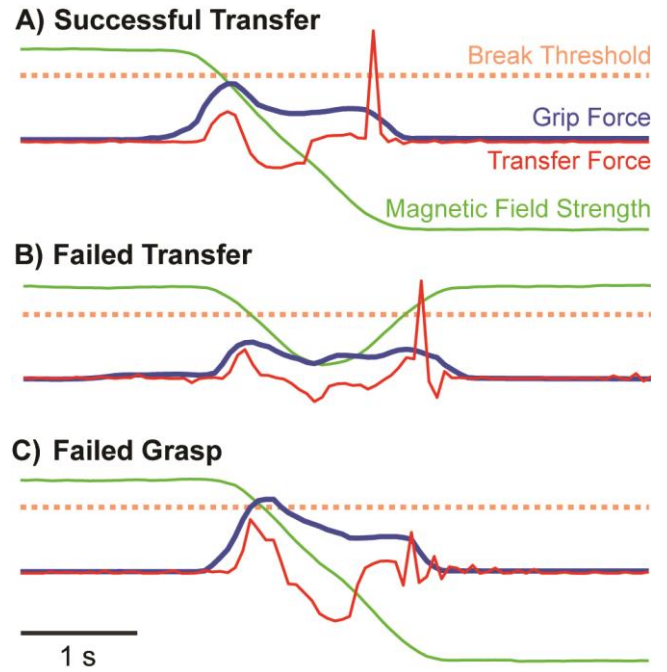


Figure 2. Example data from the EGG during gentle grasp condition. An embedded load cell records grip force (blue) and an embedded accelerometer records transfer force (red). An embedded Hall-effect sensor records magnetic field strength (green). Participants are tasked to transfer the EGG from one side of a 3D-printed barrier to another without their applied grip force (blue) exceeding a predetermined break threshold (orange dashed). The 3D-printed barrier separates two magnets of opposite polarity. (A) In a successful transfer, the magnetic field switches polarity and the grip force stays below the break threshold. (B) A failed transfer occurs when the magnetic field never switches polarity, indicating that the user placed the EGG back down on the same side of the 3D-printed barrier that it originated from. (C) A failed grasp occurs when the grip force exceeds the break threshold. In the successful transfer (A), as time progresses you see: 1) an initial increase in grip force as the user grabs the EGG, 2) an initial decrease in magnetic field strength and increase in transfer force as the user picks up the EGG, 3) a decrease in transfer force as a user stabilizes their arm at an elevated location above the 2"-tall 3D-printed barrier, 4) a continual decrease in magnetic field strength and sustained grip force as the user transfers the EGG over the barrier, 5) a spike in the transfer force as the user places the EGG on the ground, and 6) a decrease in the grip force as the user releases the EGG.

Although not analyzed in the present work, the transfer force can be used to segment different action phases during grasping [29] and may have value in future studies.

E. Data Analysis

All data were screened for normality using the Anderson-Darling test for normality. For the implicit grasping condition, data consisted of 20 peak forces and 20 transfer durations for both the paretic and non-paretic hand of each participant. For the gentle grasping condition, in which participants could break the EGG, data consisted of the total number of breaks (out of 20) as well as 20 peak forces and 20 transfer durations. Each participant was treated as a separate case study. For each participant and each condition, peak forces and transfer durations were compared between the paretic and non-paretic hands using the Wilcoxon Rank-Sum test.

III. RESULTS

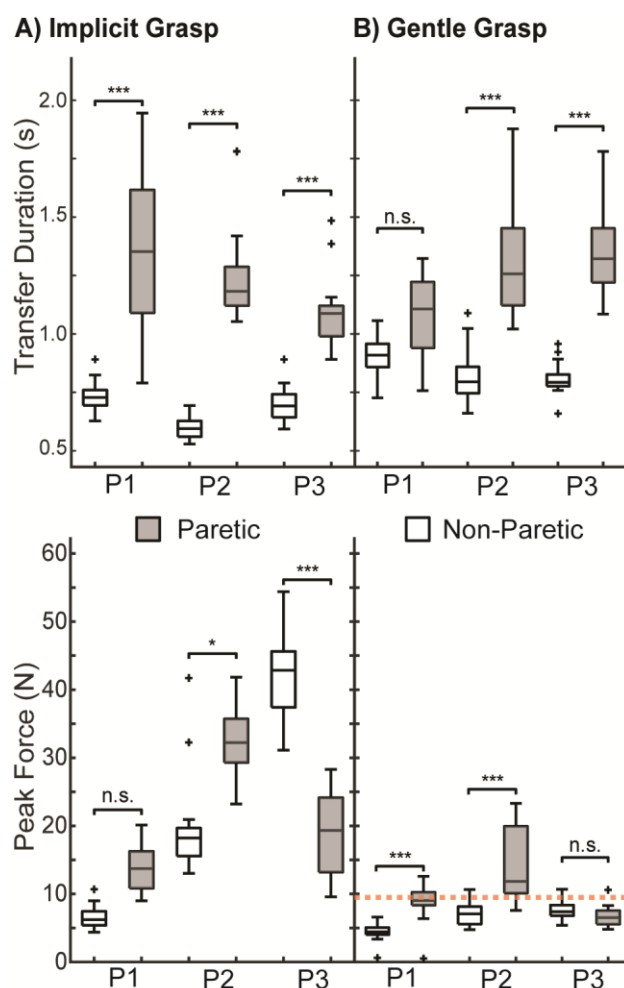


Figure 3. Transfer duration and peak force while participants attempted to transfer the EGG over a 2"-tall barrier as quickly as possible. (A) Without instruction regarding the applied grip force (implicit grasping condition), all participants took significantly longer to transfer the object with their paretic hand relative to their non-paretic hand. Interestingly, differences in implicit grasping force between the paretic and non-paretic hand were unique to each participant. (B) When tasked to minimize their grasping force to not "break" the EGG (gentle grasping condition), all participants had a harder time completing the task. P1 and P2 both exhibited significantly higher grasping forces and often broke the object. P3 did not have higher grasping forces, but did take significantly longer to move the EGG with their paretic hand. Each participant performed 20 transfers per hand per condition (N=20). * = $p < 0.05$, *** = $p < 0.001$; Wilcoxon Rank-Sum test.

A. Participants Took Longer to Transfer the EGG with their Paretic Hand during Implicit Grasping.

We tasked participants to transfer the EGG from one side of a 3D-printed barrier to another as quickly as possible. All three participants took a significantly longer time to transfer the EGG over the barrier with their paretic hand than with their non-paretic hand during implicit grasping (p 's < 0.001 ; Fig. 3A). When participants were instructed to minimize their grip force to not "break" the EGG, two of three participants took longer to transfer the EGG over the barrier with their paretic hand than with their non-paretic hand (p 's < 0.001 ; Fig. 3B).

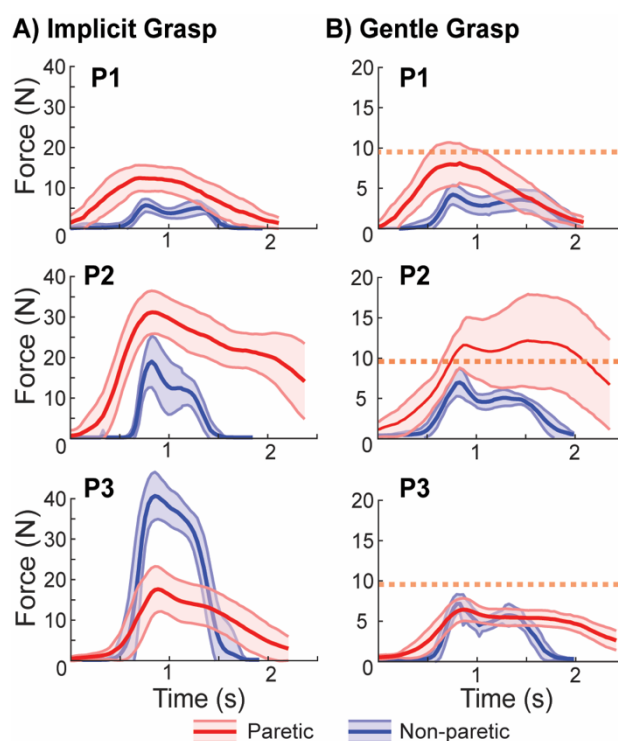


Figure 4. Example grasping forces for implicit grasping (A) and gentle grasping (B). Data show mean $\pm$ standard deviation for the paretic (red) and non-paretic (blue) hands. N = 20 transfers. Note the larger forces and force variability for the paretic hands of P1 and P2 during gentle grasping. Additionally, note the longer grasping durations for the paretic hands of P2 and P3 during gentle grasping.

B. Implicit Grasping Force Was Uniquely Different between Paretic and Non-Paretic Hands for Each Participant.

During the initial use of the EGG, participants were not instructed how much force to exert on the EGG. Thus, the force exerted represents their implicit grasping force. Interestingly, differences in implicit grasping force between the paretic and non-paretic hands were unique to each participant. P1 had no significant differences ($p = 0.11$), P2 exerted significantly greater implicit force with their paretic hand ($p < 0.05$), and P3 exerted significantly less implicit force with their paretic hand ($p < 0.001$).

C. Gentle Grasping Force Was Significantly Greater with the Paretic Hand for Two of Three Participants.

When instructed to minimize their grip force while transferring the object, participants had more difficulty completing the task. P1 exerted significantly greater force with their paretic hand than with the non-paretic hand ($p < 0.001$) and broke the EGG 9/20 times with their paretic hand compared to 0/20 times with the non-paretic hand. P2 also exerted significantly greater force with their paretic hand than with the non-paretic hand ($p < 0.001$) and broke the EGG 16/20 times with their paretic hand compared to 2/20 times with the non-paretic hand. In contrast, P3 had no significant differences between their gentle grasping forces ($p = .74$) and broke the EGG 1/20 times with their paretic hand and 3/20 times with their non-paretic hand. Examples traces of the

grasping forces are shown in Fig. 4.

IV. DISCUSSION

Our results from three chronic stroke patients highlight the patient-specific differences in sensorimotor disability after a stroke. Based on self-reports and standardized clinical assessments, P1 had considerable sensory and motor deficits, P2 and P3 had predominately motor deficits. Regardless of the type of deficit, all three participants had difficulty minimizing their grasping force during the gentle grasping condition. Participants were instructed to complete the task both as fast and as accurately as possible, and as a result participants picked different strategies favoring either speed or accuracy. P1 and P2 favored speed and ended up breaking the object more often. P3 favored accuracy and ended up taking significantly longer for each transfer. Future work can introduce additional tasks with the EGG to quantify selective performance at these extreme conditions.

Future work should also correlate various EGG tasks with clinical assessments in a larger cohort of patients and with healthy controls. Given the limited number of participants, we did not attempt to correlate the performance on the EGG tasks with the clinical and/or self-reported measures of sensory and motor function. That said, it is reasonable to assume that the implicit grasping task would predominately correlate with motor function and that the gentle grasping task would correlate with both sensory and motor function. P1 had the worst clinical scores for both sensory and motor function, and arguably performed the worst overall on both the implicit and gentle grasping tasks.

Sensory feedback and the ability to regulate grasp force play important roles in activities of daily living, such as eating delicate foods [24], [30], holding/shaking hands [24], and confidently holding various household objects [24]. In this light, future work should also correlate performance on EGG tasks with self-reported quality of life outcomes.

This work represents one of the first uses of a fragile-object task with stroke patients. Prior work with the fragile-object task involves assessing sensorimotor function in prosthesis users. [23] used a break force ratio of 0.03N/g, similar to that used in this study. Prosthesis users in [23] broke the object 37% of the time. Combined across all three stroke participants, the paretic hand broke the object 45% of the time. Despite still retaining their limb, stroke users suffer considerable sensorimotor deficits akin to that of an amputation.

The fragile-object task quantifies fine sensorimotor function during a dynamic pinch grip. In this regard, the fragile-object task is similar to the Strength-Dexterity test [31]. The Strength-Dexterity test uses compressible springs of different resistance to test an individual's ability to compress the spring without buckling the spring. The ability to compress the spring measures pinch force (i.e., strength) and the ability to prevent buckling requires sensorimotor coordination (i.e., dexterity). The EGG introduced here can also assess pinch force and sensorimotor coordination, in addition to gross motor function (i.e., transfer speed), while providing more quantitative data during a real-world transfer task with a single device. Further, the EGG could be used to

provide real-time auditory or visual feedback to users regarding their motor output.

The Strength-Dexterity test has shown that many stroke patients still have impaired precision grip force control at 6 months after stroke even in cases where the subject has good overall upper-limb and hand sensorimotor status as measured by clinical tests [32]. Ultimately, we hope the EGG can be used as a rehabilitative tool in addition to a diagnostic tool. Future work will seek to quantify the impact of repeated practice with the EGG on standardized clinical sensory and motor assessments in the stroke population. By programmatically adjusting the break threshold in between transfers, the EGG can consistently challenge and reward patients, and therefore may be able to engage patients in high-repetition sensorimotor retraining.

V. CONCLUSION

In this work, we show one of the first uses of a fragile-object task with stroke patients and introduce a novel sensing modality into the task to automate data analysis. We show that stroke patients have significant and patient-specific deficits in fine sensorimotor hand function on their paretic side. These results corroborate prior studies highlighting persistent impaired precision grip force control after stroke. The simple device and automated data segmentation introduced here may provide an avenue for patients to rehabilitate fine sensorimotor function with auditory and visual feedback. Ultimately, demonstrations of the fragile-object test in different patient populations may contribute towards more widespread quantifiable measures of sensorimotor function.

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